# Output Ontology (OO)

## Score: 1/5 — Serious concern | Confidence: high

***Benchmark:*** *MRBENCH* | ***Context:*** *Metropolitan India — Grades 1–8 Mathematics Tutoring (Student Population)*

**Output Ontology definition:** Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

Output ontology is a HIGH-priority dimension per elicitation. MRBench scores responses across eight pedagogical dimensions using a three-tier label system [Q58, Q59] aggregated into DAMR [Q64]. Critically, the deployment's primary success criterion is student satisfaction, and elicitation A4 explicitly confirms that a pedagogically correct but culturally distant response would NOT count as satisfying — yet MRBench has no output category for cultural appropriateness, NCERT curriculum correctness, or Indian classroom register. The Tutor Tone dimension trivially collapses to a non-offensiveness boundary (DAMR reaches 100% when neutral and encouraging are merged into 'Non-offensive') [Q90], failing to operationalize the warmth/encouragement register required by the deployment. The authors explicitly acknowledge the taxonomy 'does not consider the tutoring dialogues' impact on overall student learning' [Q120, Q121] and treats dimensions as orthogonal/independent [Q30, Q86] despite real interdependencies [Q116]. A response could therefore receive a high DAMR score while being culturally unsuitable for metropolitan Indian students, with no flag in the output schema. Web search confirms no validation study links MRBench scores to Indian student satisfaction [WEB-9, WEB-17, WEB-18].

| ID | Evidence |
| --- | --- |
| Q30 | "Although there are inherent interdependencies among the proposed dimensions of the taxonomy (e.g., a response that reveals the answer is less likely to be actionable, and vice versa), we explicitly instructed all annotators to treat each dimension as independent and orthogonal to minimize confounding factors and potential biases during the annotation process." (p.4) |
| Q58 | "Each dimension was annotated using a three-tier labeling system (see Figure 1 and Table 4)." (p.6) |
| Q59 | "For instance, the 'mistake identification' dimension employed the following labels: (i) yes, (ii) to some extent, and (iii) no." (p.6) |
| Q64 | "We utilize two key metrics to quantitatively assess the pedagogical effectiveness of LLMs and for comparative analysis: (1) Desired Annotation Match Rate (DAMR): This metric quantifies the percentage of responses from each human or LLM-based tutor that received the desired annotation labels." (p.6) |
| Q86 | "This further demonstrates the need to treat the dimensions as independent." (p.8) |
| Q90 | "When we combine these two labels into "Non-offensive", the DAMR score reaches 100% as we observe no offensive responses from any LLMs or humans." (p.8) |
| Q116 | "However, in practice, these dimensions may be inherently interrelated and may influence one another." (p.9) |
| Q120 | "However, one of the limitations is that it does not consider the tutoring dialogues' impact on the overall student learning." (p.9) |
| Q121 | "Specifically, the annotation pertains to the individual tutor turns within educational dialogues, which restricts our understanding of broader implications on student learning processes and learning gains, typically observed after a conversation concludes." (p.9) |
| WEB-9 | LLM-in-education meta-analysis documents motivational effects but not for Indian school-age math (https://arxiv.org/html/2507.02180v1) |
| WEB-17 | BEA 2025 Shared Task extended MRBench without Indian rater validation (https://arxiv.org/html/2507.10579v1) |
| WEB-18 | NAACL 2025 BEA proceedings: no participating system documented Indian adaptation (https://aclanthology.org/2025.naacl-long.57.pdf) |

## Strengths

- Eight orthogonal pedagogical dimensions provide fine-grained per-behavior signal that supports targeted diagnosis (e.g., GPT-4 reveals answer 47% of time) [Q69]

- Three-tier labeling system [Q58, Q59] captures partial credit for ambiguous cases
- Explicit 'desired label' for each dimension makes the decision rule transparent and auditable [Q65, Q128]
- Domain-agnostic NLG metrics (BLEU, BERTScore) are explicitly rejected for being inappropriate to pedagogical evaluation [Q4, Q7, Q8, Q10] — a defensible methodological choice

| ID | Evidence |
|---|---|
| Q4 | “General domain-agnostic natural language generation (NLG) metrics (Lin, 2004; Popovic´, 2017; Post, 2018; Gao et al., 2020; Liu et al., 2023) are not well-suited for this context, as most of them fail to account for pedagogical values and require gold references, which are often not available, especially in online interactions.” (p.1) |
| Q7 | “General domain-agnostic natural language generation (NLG) metrics like BLEU (Papineni et al., 2002), BERTScore (Lin, 2004), DialogRPT (Gao et al., 2020), and so on have been used as proxies to measure the coherence and human-likeness of AI tutor responses.” (p.2) |
| Q8 | “However, these metrics do not account for pedagogical values (Jurenka et al., 2024; Liu et al., 2024) and often require a ground truth answer to evaluate matching responses.” (p.2) |
| Q10 | “Additionally, these metrics can be easily manipulated; for instance, simple responses like "Hello" or "teacher:" (Baladón et al., 2023; Jurenka et al., 2024) can inflate scores.” (p.2) |
| Q58 | “Each dimension was annotated using a three-tier labeling system (see Figure 1 and Table 4).” (p.6) |
| Q59 | “For instance, the 'mistake identification' dimension employed the following labels: (i) yes, (ii) to some extent, and (iii) no.” (p.6) |
| Q65 | “The desired labels for each dimension are detailed in Table 2.” (p.6) |
| Q69 | “Both these LLMs perform well in identifying students' mistakes and their exact location, with Llama-3.1-405B having a slight edge as GPT-4 reveals the answer approximately 47% of the time, making its responses less actionable and impacting student's learning experience.” (p.7) |
| Q128 | “The definitions, associated labels, and the desired labels for each dimension of the proposed taxonomy are provided in Table 4.” (p.12) |

## Checklist

### Checklist item definitions:

| ID | Assessment Question |
|---|---|
| OO-1 | Review output label categories for regional relevance |
| OO-2 | Identify missing regionally specific categories |
| OO-3 | Flag categories encoding non-regional values |
| OO-4 | Consider stakeholder-driven taxonomy redesign |
| OO-5 | Document taxonomy issues harming structural/content validity |

### Checklist responses:

**OO-1**: The eight output categories (mistake identification, mistake location, guidance, actionability, coherence, tone, human-likeness, revealing the answer) are pedagogically relevant for any tutoring context but are insufficient for the Indian deployment, which additionally requires cultural appropriateness and curriculum alignment categories (A4, A1).

**OO-2**: Missing categories: (a) cultural-register/warmth specific to Indian classroom norms; (b) NCERT curriculum correctness; (c) Indian numeral convention compliance; (d) student satisfaction proxy. Tutor Tone is operationalized as non-offensiveness [Q90], which does not capture warmth required by elicitation A4.

**OO-3**: Tutor Tone's collapse to 'Non-offensive' achieving 100% DAMR [Q90] encodes a Western default of register adequacy that does not match Indian classroom warmth norms (per A4). The 'human-likeness' dimension is similarly under-specified for cultural register.

**OO-4**: Stakeholder-driven taxonomy redesign is needed: at minimum, adding a cultural-appropriateness dimension and a curriculum-alignment dimension, with desired labels validated against Indian educators and students.

**OO-5**: Documented harms: structural validity is harmed because the deployment's success construct (student satisfaction including cultural fit) has structure not captured by the eight dimensions; content validity is harmed by missing categories;

external validity is harmed because high DAMR scores cannot be assumed to predict deployment-context satisfaction. Authors themselves acknowledge taxonomy does not address overall learning impact [Q120, Q121].

| ID | Evidence |
|---|---|
| Q14 | "The evaluation taxonomy detailed in Section 4 assesses the appropriateness of the Tt+1 response across eight key pedagogical dimensions." (p.3) |
| Q64 | "We utilize two key metrics to quantitatively assess the pedagogical effectiveness of LLMs and for comparative analysis: (1) Desired Annotation Match Rate (DAMR): This metric quantifies the percentage of responses from each human or LLM-based tutor that received the desired annotation labels." (p.6) |
| Q89 | "Our findings on the Tutor Tone align with those of Wang et al. (2024a) – in task-oriented conversations, AI tutors tend to be more Neutral than Encouraging." (p.8) |
| Q90 | "When we combine these two labels into "Non-offensive", the DAMR score reaches 100% as we observe no offensive responses from any LLMs or humans." (p.8) |
| Q116 | "However, in practice, these dimensions may be inherently interrelated and may influence one another." (p.9) |
| Q118 | "We acknowledge that the proposed taxonomy will need to be verified on and likely adapted if applied to other tasks such as concept learning, and to subjects other than mathematics." (p.9) |
| Q119 | "The current taxonomy and annotation scheme focus on the appropriateness of the tutor responses." (p.9) |
| Q120 | "However, one of the limitations is that it does not consider the tutoring dialogues' impact on the overall student learning." (p.9) |
| Q121 | "Specifically, the annotation pertains to the individual tutor turns within educational dialogues, which restricts our understanding of broader implications on student learning processes and learning gains, typically observed after a conversation concludes." (p.9) |
| WEB-9 | LLM-in-education meta-analysis documents motivational effects but not for Indian school-age math (https://arxiv.org/html/2507.02180v1) |
| WEB-17 | BEA 2025 Shared Task extended MRBench without Indian rater validation (https://arxiv.org/html/2507.10579v1) |

## Information Gaps

- No published correlation between DAMR scores and student satisfaction for any population
- No empirical specification of what Indian-classroom 'warmth register' looks like at the response level

## Requires Expert Verification

- Definition and operationalization of a cultural-appropriateness output dimension by Indian educators and students
- Calibration of desired labels for Tutor Tone against Indian metropolitan student preferences

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## Remediation

**Gap 1: Tutor Tone dimension collapses to non-offensiveness [Q90]; no output category captures cultural appropriateness or student satisfaction, both confirmed essential by elicitation A4.**

**Recommendation:** Replace or supplement the Tutor Tone dimension with a multi-level cultural-register dimension calibrated to Indian classroom warmth norms; add a proxy output dimension for predicted student satisfaction validated against in-deployment engagement signals.

| ID | Evidence |
|---|---|
| Q90 | "When we combine these two labels into "Non-offensive", the DAMR score reaches 100% as we observe no offensive responses from any LLMs or humans." (p.8) |